Replication Study : Kaitlin Cox

Oil, Islam, and Women
By: Michael L. Ross

Source: The American Political Science Review,
Feb., 2008, Vol. 102, No. 1 (Feb., 2008), pp. 107-123
Published by: American Political Science Association
Stable URL: https://www.jstor.org/stable/27644501

The Original Paper: Oil, Islam, and Women

Ross (2008)

Previous research:

- Attributed gender inequality in the Middle East to Islam

This paper:

- Proposes that it is due to oil

Hypotheses:

- H_1: A rise in the value of oil production will reduce female participation in the labor force.

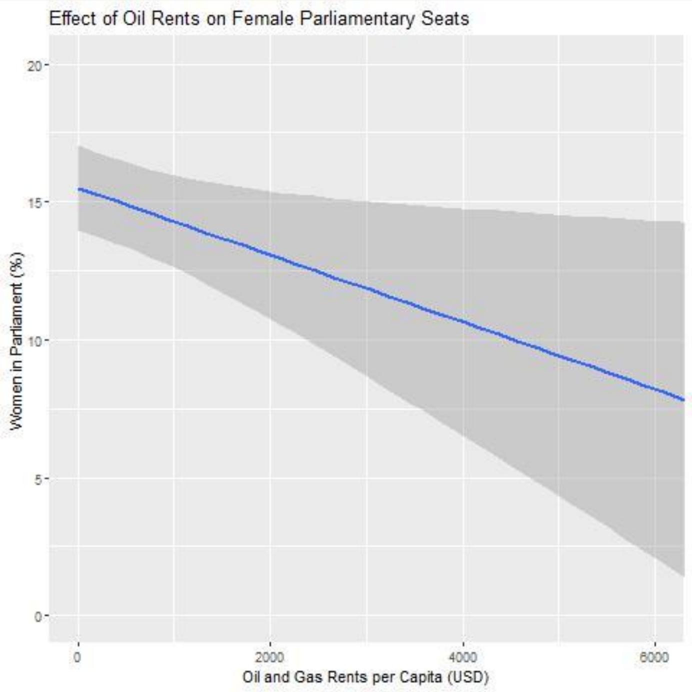
- H_2: A rise in the value of oil production will reduce female political influence.

Theory

- Ross believes that low-wage export-oriented industries, especially textiles, help women enter the labor force
- Dutch Disease models suggest that oil booms lead to less labor force participation for women
- Labor force participation is connected to social and political inclusion

The Original Paper: Findings

Oil Rents were significant in predicting both Female Labor Force Participation and Female Political Participation



The Original Paper: Female Labor Force Participation

TABLE 1. Pooled Time-Series Cross-national Regressions, with First Differences and Fixed Effects

Dependent variable is Female Labor Force Participation, 1960–2002				
	(1)	(2)	(3)	(4)
Income (log)	−0.011	−0.039	−0.014	−0.051
	(0.36)	(1.19)	(0.53)	(1.08)
Income squared (log)	0.017	0.049	0.021	0.021
	(0.52)	(1.46)	(0.75)	(0.43)
Working Age	0.115	0.115	0.066	0.177
	(4.67)***	(4.66)***	(2.73)**	(13.65)***
Oil Rents per capita		−0.026	−0.017	−0.049
		(4.02)***	(2.34)*	(4.33)***
Observations	5234	5234	5168	5395
Countries	161	161	159	161
R-squared: within	.005	.008	.003	.05
R-squared: between	.16	.18	.19	.22

Note: Absolute value of t statistics in parentheses. *Significant at 5%; **significant at 1%; ***significant at 0.1%. Country fixed-effects are used in each estimation. In column 3, the two most influential countries have been dropped from the sample. In column 4, year dummies were included in place of the AR1 process.

The Original Paper: Female Seats in Parliament

TABLE 4. Cross-national Regressions on Female Seats in Parliament

	Dependent variable is parliamentary seats held by women (%), 2002							
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Income (log)	0.311	0.259	0.364	0.320	0.329	0.39	0.367	0.779
	(3.80)***	(2.81)**	(4.39)***	(3.41)***	(3.74)***	(4.13)***	(4.13)***	(5.68)***
Middle East	−0.378	−0.261	−0.278	−0.193	−0.08	−0.233	−0.253	−0.182
	(6.07)***	(3.09)**	(4.62)***	(2.42)*	(0.84)	(2.82)**	(3.28)***	(1.28)
Islam		−0.178		−0.139	−0.103	−0.18	−0.144	−0.123
		(1.93)		(1.56)	(1.17)	(1.85)	(1.50)	(0.71)
Oil Rents per capita			−0.232	−0.218	−0.152	−0.258	−0.238	−0.317
			(3.32)***	(3.32)***	(2.16)*	(3.40)***	(3.46)***	(3.64)***
Polity						−0.158	−0.298	−0.294
						(1.54)	(2.94)**	(1.97)
Proportional Representation							0.316	0.013
							(4.30)***	(0.10)
Closed List								0.269
								(2.94)**
Female Labor Force Participation					0.309			
					(3.68)***			
Observations	161	161	161	161	160	161	161	88
R-squared	0.21	0.23	0.25	0.26	0.33	0.27	0.35	0.40

Note: Robust t statistics in parentheses. All variables are standardized. *Significant at 5%; **significant at 1%; ***significant at 0.1%.

The Original Paper: Female Ministerial Positions

TABLE 5. Cross-national Regressions on Female Ministerial Positions 2002

	Dependent variable is ministerial seats held by women (%), 2002				
	(1)	(2)	(3)	(4)	(5)
Income (log)	0.250	0.239	0.282	0.278	0.281
	(2.82)**	(2.50)*	(3.01)**	(2.73)**	(2.76)**
Middle East	−0.244	−0.218	−0.181	−0.174	−0.141
	(4.94)***	(3.21)**	(4.76)***	(2.98)**	(2.31)*
Islam		−0.039		−0.012	−0.002
		(0.49)		(0.14)	(0.03)
Oil Rents per capita			−0.139	−0.138	−0.117
			(2.30)*	(2.23)*	(1.83)
Female Labor Force Participation					0.094
					(1.26)
Observations	155	155	155	155	154
R-squared	0.11	0.11	0.12	0.12	0.12

Note: Robust t statistics in parentheses. All variables are standardized. *Significant at 5%; **significant at 1%; ***significant at 0.1%.

My Replication

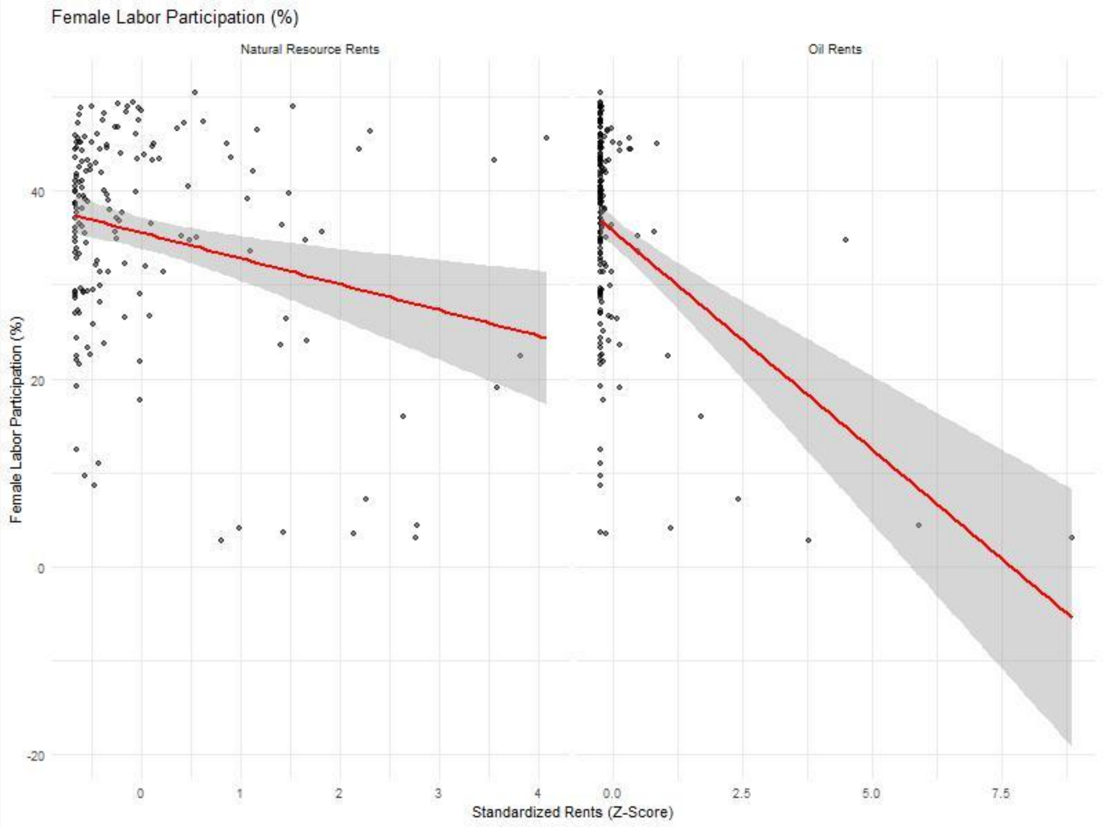
My Contribution:

- Total Natural Resource Rents as the outcome variable
- Theorizing that that the impact of natural resources on female inclusion may extend beyond oil

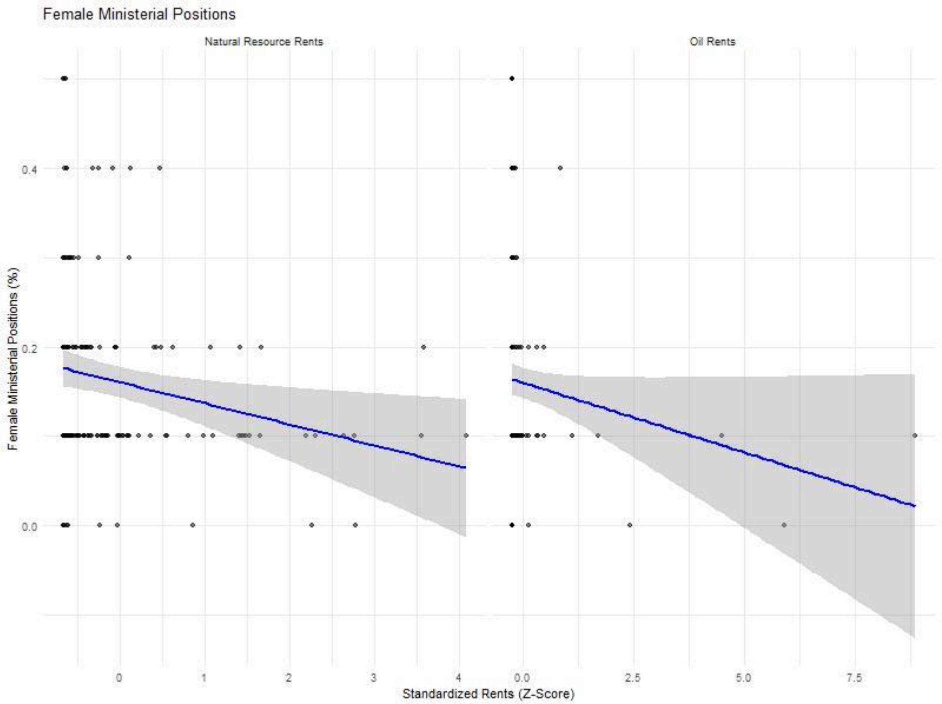
My Results:

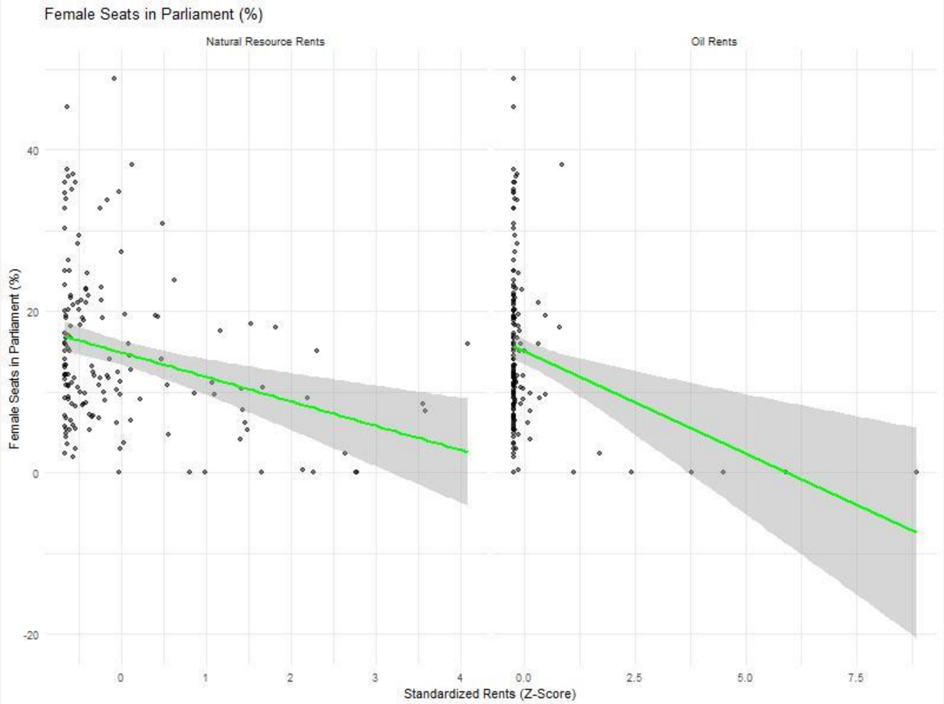
- Not significant for Ministerial Seats Held by Women or Female Labor Force Participation
- Significant for Parliamentary Seats Held by Women (but not as significant as Oil Rents)

Oil has a stronger negative association with Female Labor Participation than Natural Resource Rents has



However, the difference is less dramatic for the % of Female Ministers and % of Female Seats in Parliament





Model Comparison: Female Labor Force

Oil Rents per Capita: -0.210 (significant at 0.01)

Natural Resource Rents per Capita: -0.070 (not significant)

	Dependent variable:	
	Female Labor Force Participation	
	(1)	(2)
Income (log)	−1.864***	−1.512**
	(0.654)	(0.678)
Income squared (log)	2.122***	1.765***
	(0.648)	(0.669)
Working Age	−0.350***	−0.428***
	(0.119)	(0.126)
Middle East	−0.326***	−0.378***
	(0.090)	(0.093)
Communist	0.286***	0.307***
	(0.082)	(0.086)
Islam	−0.139	−0.164*
	(0.086)	(0.089)
Oil Rents per Capita	−0.210***	
	(0.072)	
Natural Resource Rents per Capita		−0.070
		(0.074)
Constant	−0.012	−0.013
	(0.060)	(0.062)
Observations	167	163
R^2	0.427	0.402
Adjusted R^2	0.401	0.375
Residual Std. Error	0.776 (df = 159)	0.796 (df = 155)
F Statistic	16.898*** (df = 7; 159)	14.912*** (df = 7; 155)
Note:		*p<0.1; **p<0.05; ***p<0.01

Model Comparison: Parliamentary Seats Held by Women

Oil Rents per Capita: -0.259
(significant at 0.01)

Natural Resource Rents per Capita: -0.170
(significant at 0.05)

	Dependent variable:	
	Parliamentary Seats Held by Women (percent), 2002	
	(1)	(2)
Income (log)	−1.748**	−1.481*
	(0.733)	(0.765)
Income squared (log)	2.163***	1.894**
	(0.728)	(0.756)
Working Age	−0.090	−0.218
	(0.134)	(0.144)
Middle East	−0.122	−0.160
	(0.100)	(0.102)
Communist	0.140	0.194*
	(0.095)	(0.100)
Islam	−0.147	−0.167
	(0.100)	(0.104)
Oil Rents per Capita	−0.259***	
	(0.079)	
Natural Resource Rents per Capita		−0.170**
		(0.085)
Constant	−0.015	−0.011
	(0.068)	(0.071)
Observations	160	156
R^2	0.305	0.278
Adjusted R^2	0.273	0.244
Residual Std. Error	0.854 (df = 152)	0.879 (df = 148)
F Statistic	9.513*** (df = 7; 152)	8.128*** (df = 7; 148)
Note:		*p<0.1; **p<0.05; ***p<0.01

Model Comparison: Ministerial Seats Held by Women

Oil Rents per Capita: -0.145
(not significant)

Natural Resource Rents per Capita: -0.145
(not significant)

	Dependent variable:	
	Ministerial Seats Held by Women (percent) 2002	
	(1)	(2)
Income (log)	−0.516	−0.446
	(0.810)	(0.825)
Income squared (log)	0.894	0.817
	(0.803)	(0.815)
Working Age	−0.151	−0.227
	(0.154)	(0.158)
Middle East	−0.171	−0.179
	(0.118)	(0.117)
Communist	−0.110	−0.070
	(0.107)	(0.109)
Islam	−0.029	−0.036
	(0.120)	(0.122)
Oil Rents per Capita	−0.145	
	(0.088)	
Natural Resource Rents per Capita		−0.145
		(0.093)
Constant	−0.010	−0.005
	(0.076)	(0.077)
Observations	154	151
R^2	0.174	0.165
Adjusted R^2	0.134	0.124
Residual Std. Error	0.933 (df = 146)	0.939 (df = 143)
F Statistic	4.383*** (df = 7; 146)	4.042*** (df = 7; 143)
Note:		*p<0.1; **p<0.05; ***p<0.01

Conclusion

- Further research is needed to see if the results would hold in other studies (including with current data)

If the findings of Ross's study and my replication hold:

- There is a correlation between female labor force and political participation and oil that is different from that of other natural resources